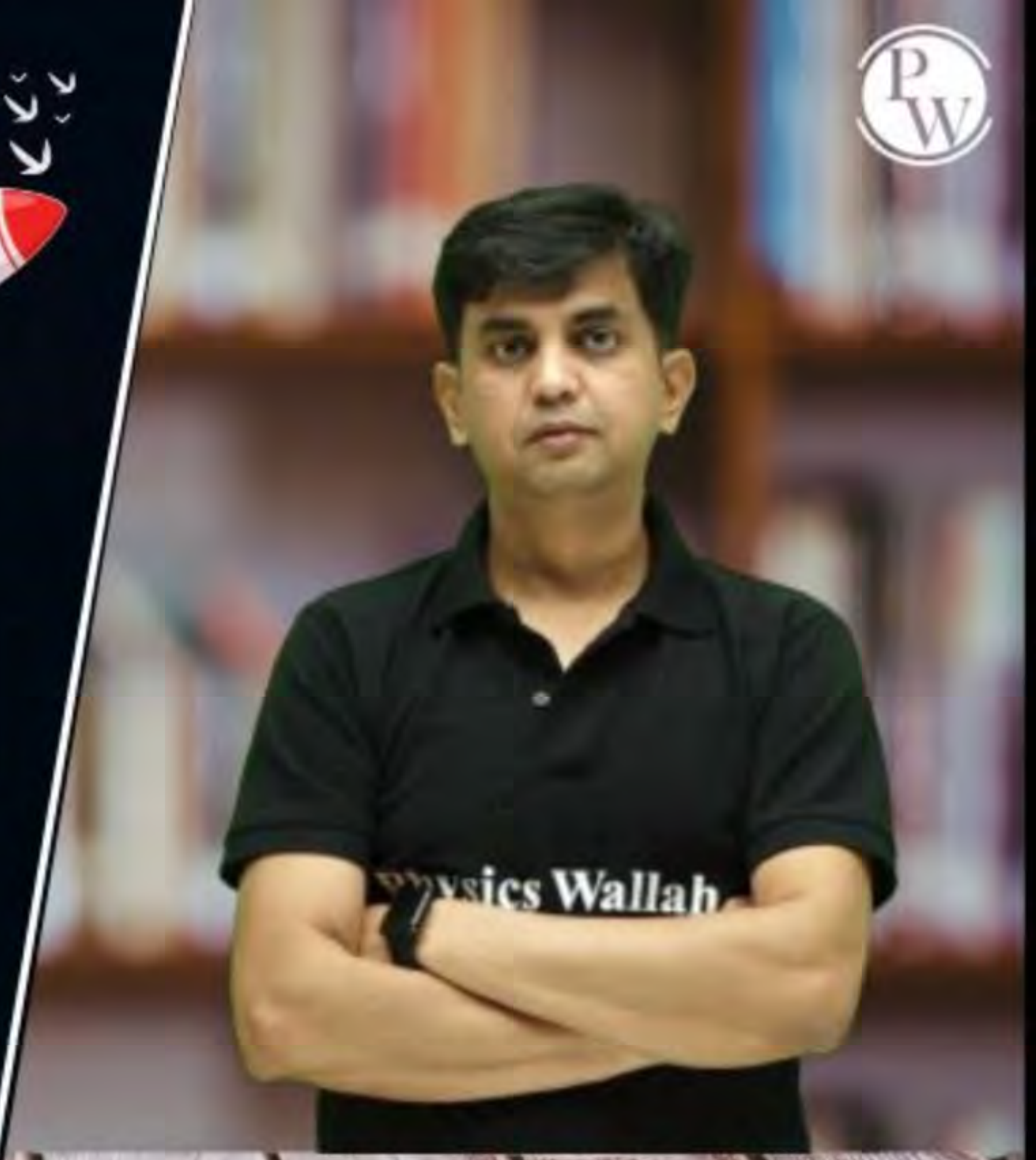
CS & IT ENGINEERING

Computer Network - 2

Network Layer

Lecture No. - 01



By - Abhishek Sir

Recap of Previous Lecture

Topic Hamming Code

Topic Hamming Distance

Topic Optimum Packet Size

Topics to be Covered

Topic Identification Number CS-14

Topic IPv4 Options [CS-17]

Topic ICMP

[MCQ] [GATE-2021][2 Mark]

#Q. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits			
c_8	c_4	c_2	c_1
y	0	1	0

Which of the following choices gives the correct values of x and y?

(A) x is 0 and y is 0

(B) x is 0 and y is 1

(C) x is 1 and y is 0

(D) x is 1 and y is 1

$\boxed{\text{Ans: A}}$

→ Even/odd Parity → Even Parity

1	1	0	X	Y	0	1	0	0	1	1	0
12	11	10	9	8	7	6	5	4	3	2	1

$\boxed{0}$

$\boxed{X = 0}$

$\boxed{Y = 0}$

Solution :-

1	1	0	X	**Y**	0	1	0	**0**	1	**1**	**0**
12	11	10	9	8	7	6	5	4	3	2	1

=> Identify even or odd parity

→ Parity $\mathbf{R_1}$ is balanced using "even parity"

=> Identify value of variable X

→ As per parity $\mathbf{R_0}$, X should be "zero bit"

=> Identify value of variable Y

→ If X is "zero", then as per parity $\mathbf{R_3}$, Y should be "zero bit"

[GATE-1992]

#Q. Consider a 3-bit error detection and 1-bit error correction hamming code for 4-bit data. The extra parity bits required would be <u>01</u> and the 3-bit error detection is possible because the code has a minimum distance of <u>04</u>.

$\boxed{\text{Ans: 1 \& 4}}$

Hamming Code (N, m, d)

① Base method ⇒ $\boxed{d=3}$
→ Two-bit Error detected;
→ one-bit Error Corrected;

② Modified Method ⇒ $\boxed{d=4}$
→ Three-bit Error detect;
→ one-bit Error corrected;

d_3	d_2	d_1	R_2	d_0	R_1	R_0	P
7	6	5	4	3	2	1	

Topic : Hamming Code

→ Even Parity

d_3	d_2	d_1	R_2	d_0	R_1	R_0	P
7	6	5	4	3	2	1	Parity

$R_0 = d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6$ [1 = 3, 5, 7]

$R_1 = d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6$ [2 = 3, 6, 7]

$R_2 = d_1 \oplus d_2 \oplus d_3 \oplus d_7$ [4 = 5, 6, 7]

$P = R_0 \oplus R_1 \oplus d_0 \oplus R_2 \oplus d_1 \oplus d_2 \oplus d_3$

Topic : Hamming Code → Linear Code

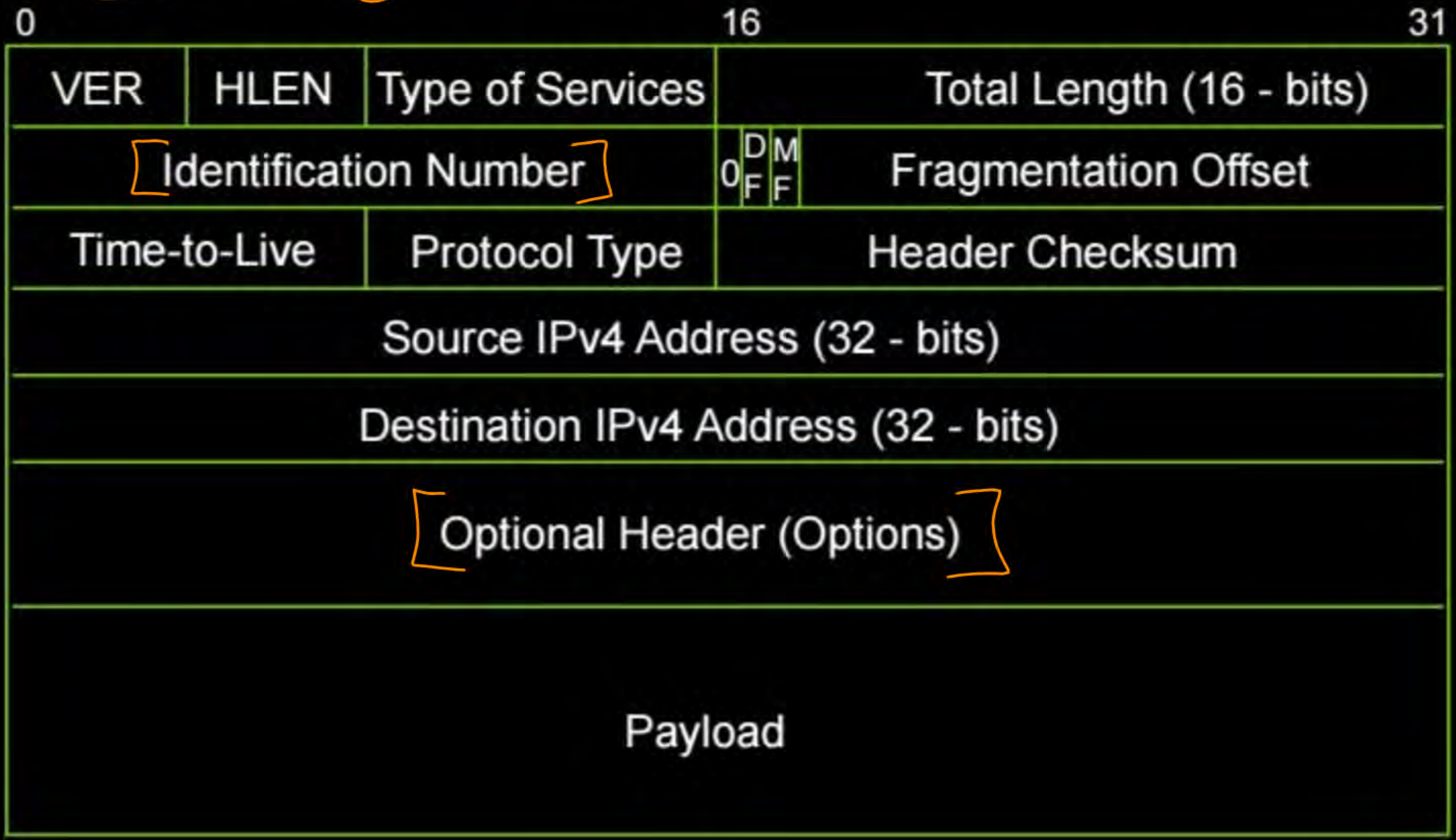
Hamming Code (with Even Parity) and 3 data bits and One More Parity Bit :

Data		Codeword
$d_2d_1d_0$	→	$d_2d_1C_2d_0C_1C_0P$
0 0 0	→	0 0 0 0 0 0 0
0 0 1	→	0 0 0 1 1 1 1
0 1 0	→	0 1 1 0 0 1 1
0 1 1	→	0 1 1 1 1 0 0
1 0 0	→	1 0 1 0 1 0 1
1 0 1	→	1 0 1 1 0 1 0
1 1 0	→	1 1 0 0 1 1 0
1 1 1	→	1 1 0 1 0 0 1

$\boxed{2^3}$

min^m H.D. = 4

Topic : IPv4 Packet Header



Topic : MTU

→ Maximum Transmission Unit [MTU]

→ Measurement in bytes

→ Size of largest PDU (IP datagram) that can be communicated over a network

Frame [IP Datagram]

Topic : MTU

H_S
Source Host
R_S
Source Router
H_D
Destination Host
R_D
Destination Router
Source Network
Destination Network

Topic : MTU

→ Different networking technologies may have different MTU

→ MTUs for common media / static MTU (in bytes)

1. Ethernet v2	:	$\boxed{1500}$
2. Wi-Fi (WLAN, IEEE 802.11)	:	2304
3. Token Ring (IEEE 802.5)	:	4464
4. FDDI	:	4352

⇒ Dynamic MTU

Topic : MTU

→ Source host creates IPv4 datagram as per source network MTU

→ At intermediate IPv4 router, for an received IPv4 datagram if IPv4 datagram size is greater then next network (link) MTU size then need to do fragmentation according to MTU

Topic : Identification Number

→ 16 bits long $\Rightarrow$ Wrap-around $[0 \text{ to } 2^{16}-1]$

→ Assigned by source host only
[Assigned unique identification number to each transport layer Segment]

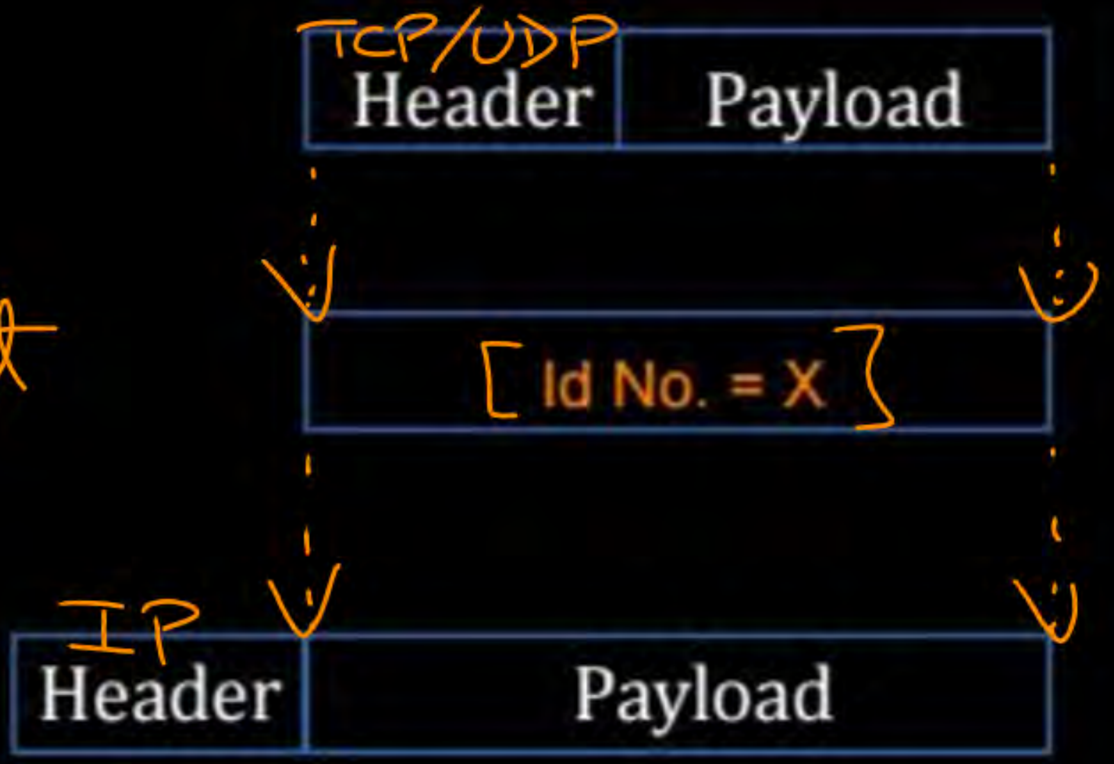
→ IP fragments of same segment, must have same identification number
[does not change during routing]

Topic : Identification Number

AT Source Host :-

Transport Layer PDU (Segment)

SDU for Network Layer = segment



Id No. = X

offset = 0

M = 0

Topic : Identification Number

AT Source Host :-

Transport Layer PDU (Segment)

TCP/UDP

Header	Payload

SDU for Network Layer

[Id No. = Y]

Header	Payload

Id No. = Y

offset = 0

M = 1

Header	Payload

Id No. = Y

M = 1

Header	Payload

Id No. = Y

M = 0

Host A

segment

Id No. = Y

[IP D$_1$, D$_2$, D$_3$]

Host B

$\underbrace{\text{IP D}_1, \text{D}_2, \text{D}_3}_{\text{ID No.} = Y}$

$\underbrace{\text{IP D}_1, \text{D}_2}_{\text{ID No.} = Y}$

Host C

segment

Id No. = Y

[IP D$_1$, D$_2$]

Topic : Identification Number

→ Receiver host do clustering of fragments, before assembly

→ based on Source IP Address and Identification number

Identification Number (16 bits) : Locally unique

Source IP Address & Identification Number (48 bits) : Globally unique

IPv4 ⇒ 32 bit

(16 bit)

Example 1 :-

[NAT]

#Q. Consider each host in an IPv4 network uses up to 1024 unique identifiers per second. After what period (in seconds) will the identifiers generated by a host wrap around ?

Identification Number : $\boxed{16 \text{ bits}}$

Total number of unique identifiers = $\boxed{2^{16}}$ $[0 \text{ to } 2^{16}-1]$
[In each host, before wrap-around]

1024 identifiers → 1 Sec

2^{16} identifiers → $\frac{2^{16}}{2^{10}} * 1 \text{ sec}$

→ 2^{6} sec

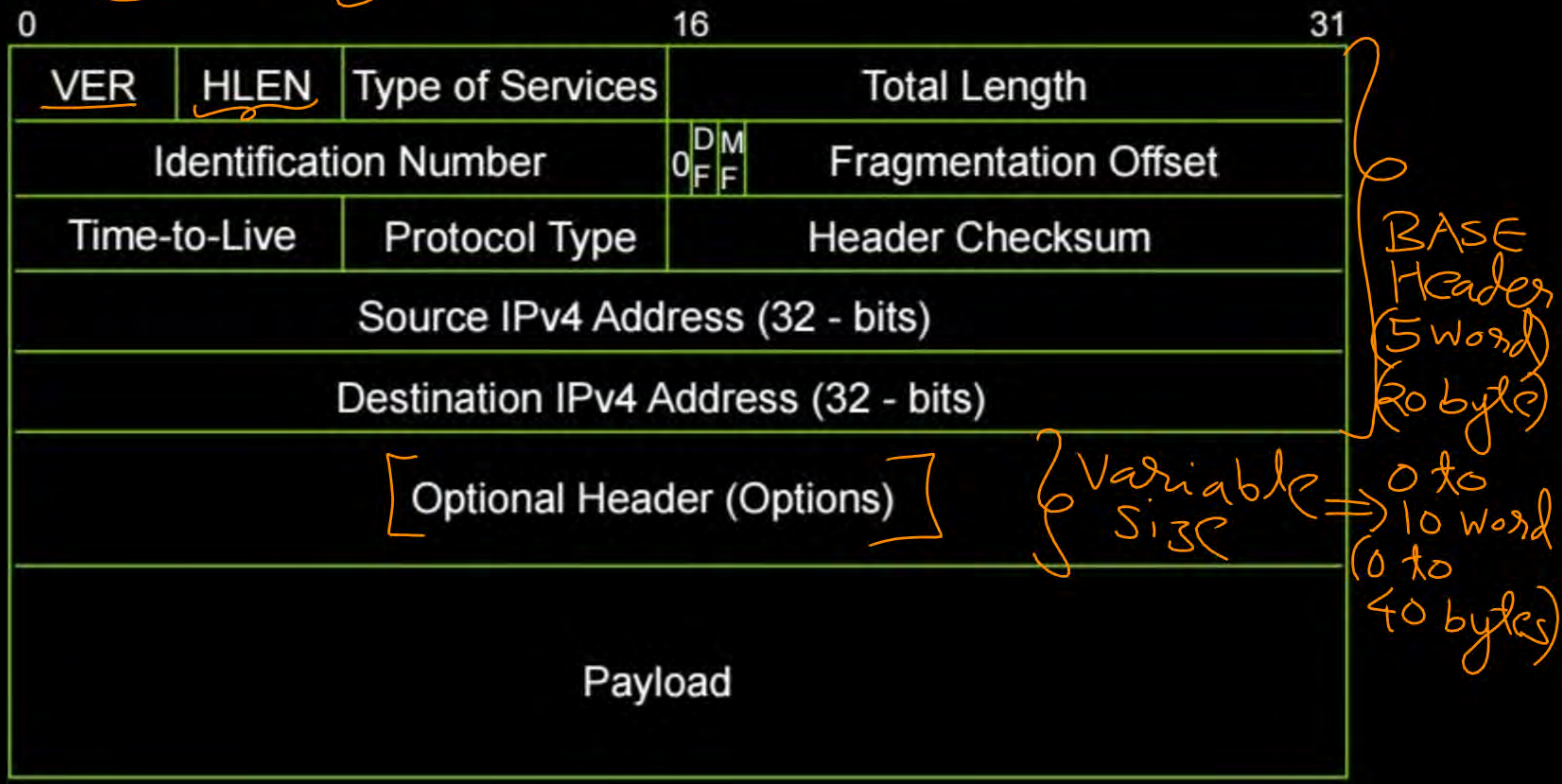
→ 64 sec

$\boxed{\text{Ans} = 64}$

[NAT] IIT-KGP, H.W. [GATE-2014] [2 Mark]

#Q. Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

Topic : IPv4 Packet Header



Topic : IPv4 Optional Header

→ Also known as "Options"

→ Options are variable in size

→ Maximum size of IPv4 optional header is 10 words [Words of 32 bits, i.e. 40 bytes]

→ if options size are not in words than it uses padding [End-of-Option Option]

Topic : IPv4 Optional Header

→ Options field component :-

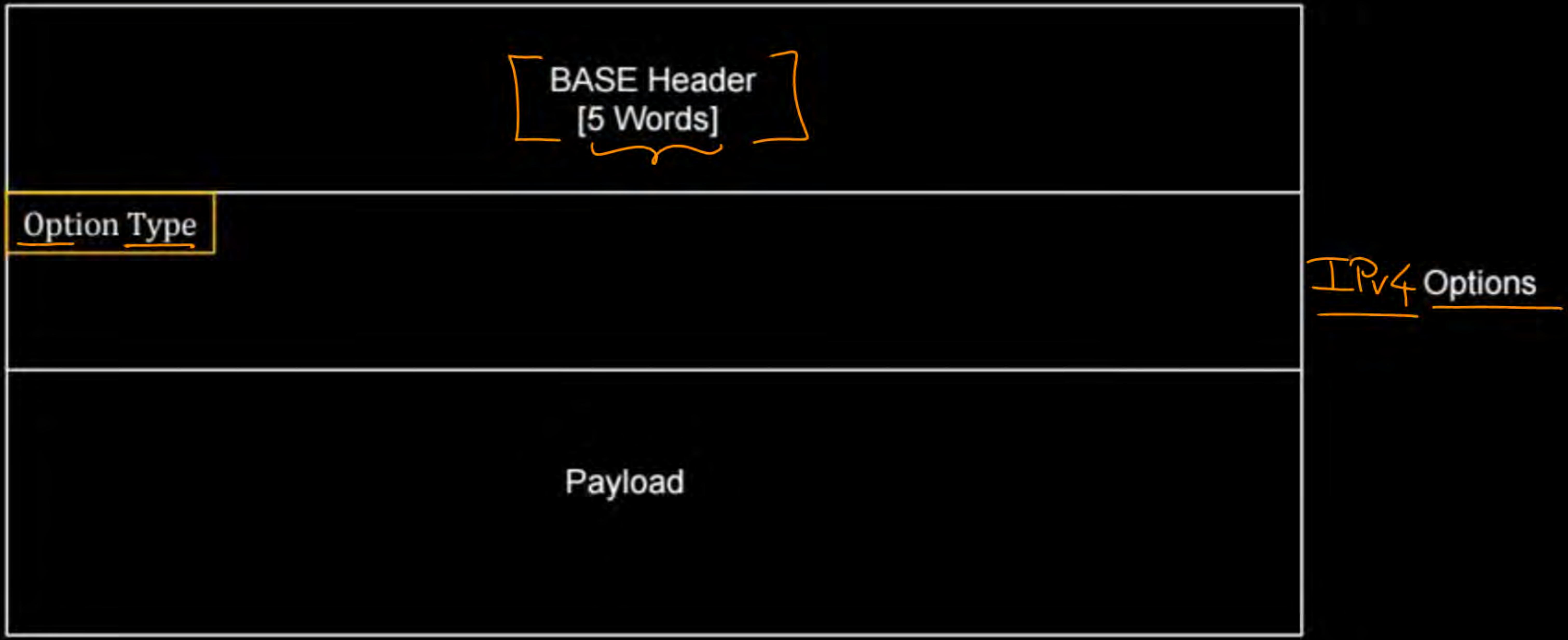
1, Option Type (1 byte) → compulsary

2. Option Length (1 byte)

3. Option Data

[Optional]

Topic : IPv4 Optional Header



Topic : Option Type

→ Option type field are of size 8-bits (1 byte) → [0 to 255]

→ Option type field are divided into three sub-fields

1. Copied	(1-bit)
2. Option Class	(2-bits)
3. Option Number	(5-bits)

Copied	Optional Class	Optional Number (5-bits)

Copied (0) : 0 to 127

Copied (1) : ~~0 to 127~~ 128 to 255

Topic : Option Type

Number	Option Type
0	End of Option [Padding]
1	No Operation
7	Record Route
68	Timestamp
82	Traceroute
131	Loose Source Route
137	Strict Source Route

Topic : Option Length

→ Option length field are of size 8-bits (1 byte)

→ Define total length of Options (in bytes)
[including option type field and option length field itself]

Topic : Record Route

→ Used to record the IP address of Internet Routers [Routers those handle the datagram along the path]

→ It can list up to "nine" router IP address

→ Source Router to Destination Router

Topic : Record Route

<table>
<tr><td>Type = 1
00000001</td><td>Type = 7
00000111</td><td>Total Length = 39
00100111</td><td>Pointer = ~~4~~ ~~8~~ 12
00000100</td></tr>
<tr><td colspan="4">First IP Address (Empty when started) (Source Router IP)</td></tr>
<tr><td colspan="4">Second IP Address (Empty when started) (R_1 IP)</td></tr>
<tr><td colspan="4">Third IP Address (Empty when started)</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td colspan="4">(9th) Last IP Address (Empty when started)</td></tr>
</table>

[NAT] IIT-R [GATE-2017][1 Mark]

#Q. The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is ________.

IPv4 optional Header = max^m 40 byte

IPv4 Address = 4 bytes

* 3 byte required by RR;

$\boxed{\text{Ans} = 9}$

Topic : Timestamp

→ Similar to the Record Route

→ Every router need to record the IP address and Timestamp

→ 32-bits Timestamp in milliseconds from UTC.

⇒ Max^m 4 Router's Entry

2 mins Summary

Topic Identification Number

Topic IPv4 Options

Topic

THANK - YOU

@ABHISHEKSIRCS_PW